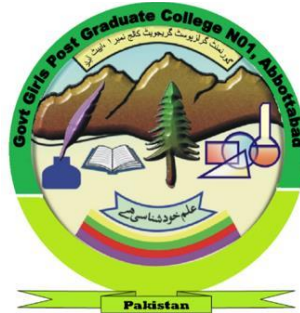


Smart Exam System

Project Report



Submitted by: Muskan Javed(23-118),
Aqsa Tariq(23-135)

Course: Visual Programming

Department: BS_CS

Semester: VI

Submitted to: Ma'am Asmara

Year: 2026

Table of Contents

Table of Contents	2
1. Introduction	3
2. Objectives.....	3
3. Tools and Technologies Used.....	3
4. System Overview.....	3
5. Functional Requirements	4
6. Non-Functional Requirements	4
7. System Modules.....	4
7.1 Splash Screen	4
7.2 Login Module	5
7.3 Registration Module	6
7.4 Student Automation – Roll Number & Onboarding.....	6
7.5 Student Dashboard	7
7.6 Exam Module	8
7.7 Admin Dashboard	9
7.8 Admin Performance Matrix Module.....	10
8. Database Design.....	12
8.1 Users Collection	12
8.2 Results Collection.....	12
9. Implementation Details	13
9.1 Registration Validation.....	13
9.2 Automatic Roll Number Generation	13
9.3 Result Calculation.....	13
9.4 Fetching the Performance Matrix.....	13
10. Testing.....	14
11. Conclusion.....	14

1. Introduction

The Smart Exam System is a cloud-based online examination and performance matrix platform built using C# .NET Windows Forms (WinForms) with MongoDB Atlas as the backend. It provides a structured and efficient way for students to register, take exams, and view their results, while administrators can manage exam records and monitor performance across the institution.

The application begins with a clean, flat-styled login and registration system, followed by role-specific dashboards. Students are guided through a fully automated onboarding flow, while administrators have a centralized matrix to track and evaluate results in real time.

2. Objectives

- Eliminate manual errors in exam marking and percentage calculation.
- Use the MongoDB Atlas cloud database to improve data security and accessibility.
- Implement automatic percentage calculation and a Pass/Fail performance matrix.
- Provide a professional, user-friendly interface for both admin and student roles.
- Automate student onboarding through unique roll number generation and default parameter assignment.

3. Tools and Technologies Used

- Programming Language: C# .NET (WinForms)
- Database: MongoDB Atlas (Cloud NoSQL)
- IDE: Visual Studio 2022
- UI Design: Flat UI Design Principles

4. System Overview

The system is designed for two types of users: Students and Admins. The application flow is as follows:

Splash Screen → Login / Registration → Student Dashboard or Admin Dashboard → Exam / Performance Matrix → Logout

Students register, are automatically assigned a roll number and starting semester, attempt exams, and view their results immediately after submission. Admins view all student records and exam results through a centralized performance matrix, where each result is automatically evaluated and color-coded as Pass or Fail. All data is stored and retrieved from the MongoDB Atlas cloud database.

5. Functional Requirements

- Role-based login and registration for Student and Admin.
- Strict age-limit validation (minimum 15 years) during registration.
- Real-time email format verification and password-match confirmation.
- Automatic roll number generation in the format NIMS-[Year]-ST[RandomID].
- Automatic assignment of default starting parameters (e.g. “1st Semester”) for new students.
- Real-time result calculation with automatic Pass/Fail status.
- Centralized Admin Performance Matrix showing live results fetched from MongoDB.
- Secure logout for both roles.

6. Non-Functional Requirements

- Clean and consistent blue-branded flat UI theme throughout all forms.
- Password masking with a Show/Hide toggle on all password fields.
- Robust exception handling to prevent crashes from data type mismatches or deserialization errors.
- Reliable cloud connectivity for fast and accurate data fetch from MongoDB Atlas.
- Role-based access control ensuring clear separation between student and admin data.

7. System Modules

Following are the system modules:

7.1 Splash Screen

Displays the application name and a short loading animation before the Login Form opens automatically.



a. Splash Screen

7.2 Login Module

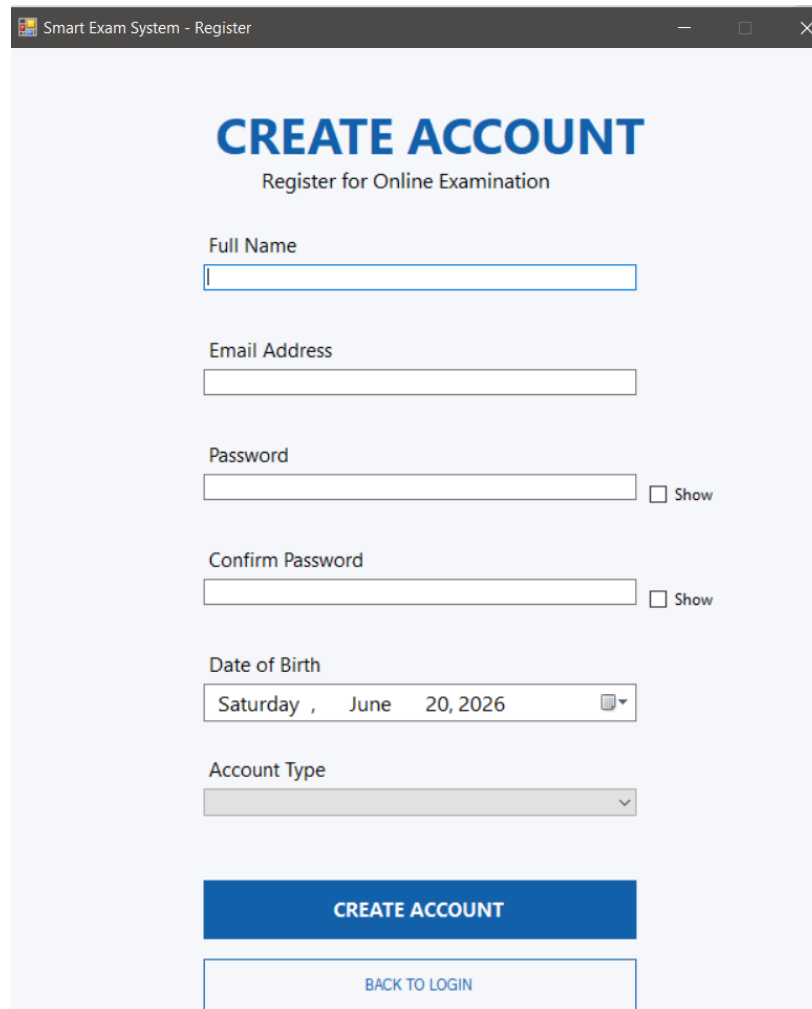
Students and Admins log in using their registered email and password. Based on their role, they are redirected to their respective dashboard. Incorrect credentials display an error message. A Show/Hide toggle allows users to view or mask their password.

The image is a screenshot of the 'Smart Exam System' login interface. The window title is 'Smart Exam System'. The interface is split into two main sections. The left section has a dark blue background and contains a white cloud icon, the text 'SMART EXAM SYSTEM' in large white letters, and 'Cloud Based Online Examination Platform' in smaller white text. At the bottom of this section, it says 'Secure • Fast • Reliable'. The right section has a light gray background and is titled 'Welcome Back' in bold black text. It contains two input fields: 'Email Address' and 'Password'. The 'Password' field has a 'Show' checkbox to its right. Below the input fields are two buttons: a blue 'LOGIN' button and a white 'CREATE ACCOUNT' button with a blue border.

b. Login Module

7.3 Registration Module

New students fill in their Name, Email, Date of Birth, and Password. The system enforces a strict minimum age of 15 years, validates the email format, and checks that the Password and Confirm Password fields match before allowing registration.



The screenshot shows a web browser window titled "Smart Exam System - Register". The main heading is "CREATE ACCOUNT" in large blue letters, with the subtitle "Register for Online Examination" below it. The form contains the following fields and controls:

- Full Name:** A text input field.
- Email Address:** A text input field.
- Password:** A text input field with a "Show" checkbox to its right.
- Confirm Password:** A text input field with a "Show" checkbox to its right.
- Date of Birth:** A date picker showing "Saturday, June 20, 2026".
- Account Type:** A dropdown menu.
- CREATE ACCOUNT:** A prominent blue button.
- BACK TO LOGIN:** A light blue button at the bottom.

c. Registration Module

7.4 Student Automation – Roll Number & Onboarding

Upon successful registration, the system automatically generates a unique institutional roll number formatted as NIMS-[Year]-ST[RandomID], and assigns the student to default starting parameters, removing the need for manual setup by an administrator.

Cluster0 > SmartExamDB > Students

Documents 1 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

ADD DATA BULK EXPORT CODE

25 1-1 of 1

_id	RollNumber	Name	Semester	Email
1	NIMS-2026-ST710	Maryam	1st Semester	may@gmail.com

d. Automatic Roll Number Generation

7.5 Student Dashboard

Displays a personalized welcome message for the logged-in student along with navigation options to attempt exams, view results, and log out.

Student Console - Smart Exam System

Roll No: NIMS-2026-ST
also@gmail.com

Available Exams

My Performance

Log Out

Your Examination Portal Dashboard

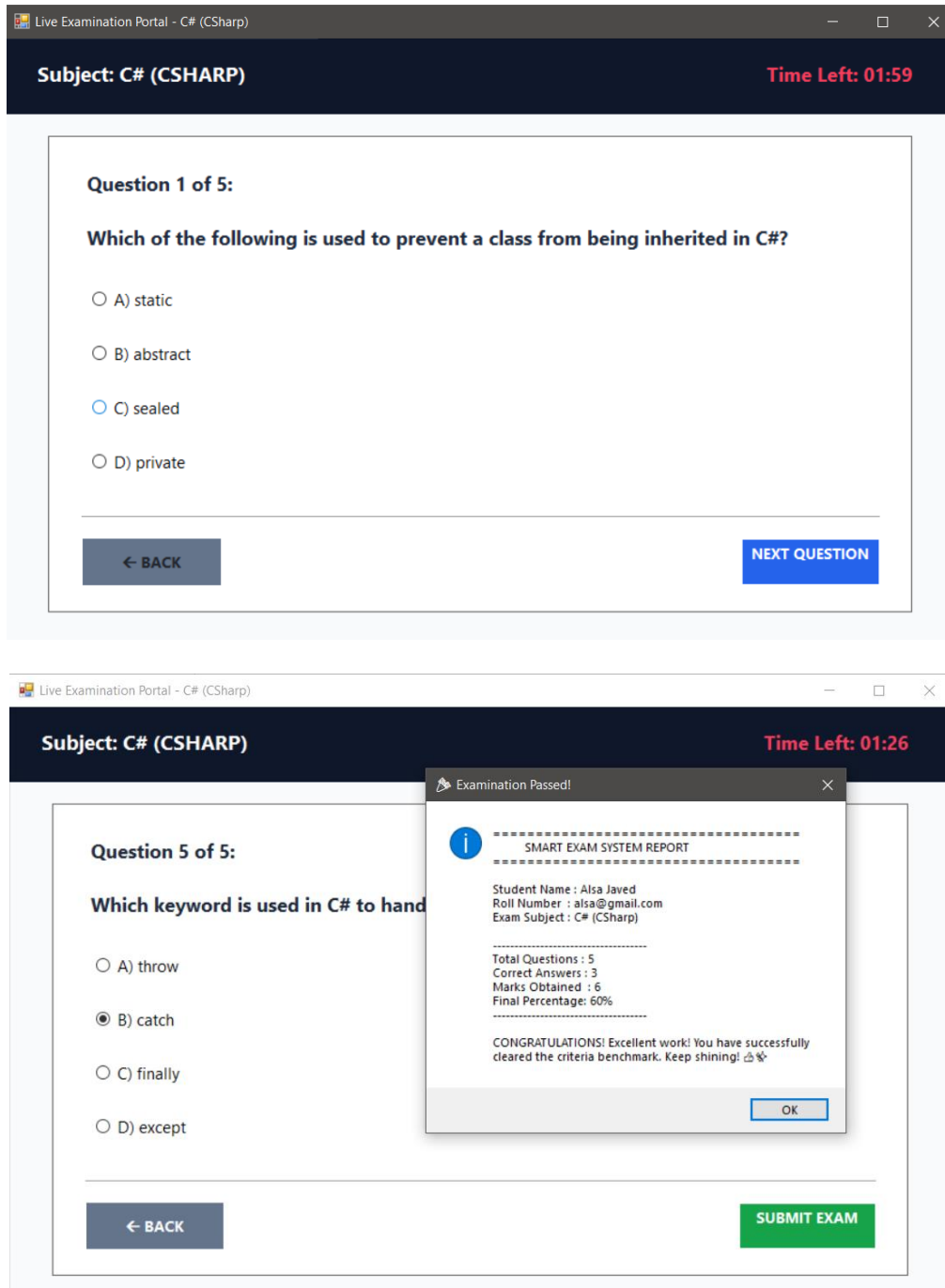
Select any assigned active assessment card below to initiate your online automated exam session.

- C# (CSHARP)**
Duration: 2 Mins | Questions Count: 5
[START EXAM](#)
- ARTIFICIAL INTELLIGENCE (AI)**
Duration: 2 Mins | Questions Count: 5
[START EXAM](#)
- NATURAL LANGUAGE PROCESSING (NLP)**
Duration: 2 Mins | Questions Count: 5
[START EXAM](#)
- ENTREPRENEURSHIP**
Duration: 2 Mins | Questions Count: 5
[START EXAM](#)
- DATA MINING**
[START EXAM](#)

e. Student Dashboard

7.6 Exam Module

Allows the student to attempt the assigned exam. On submission, the system automatically calculates the total marks and percentage and stores the result in the MongoDB Results collection.



Cluster0 > SmartExamDB > Results

Documents 7 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA BULK EXPORT CODE

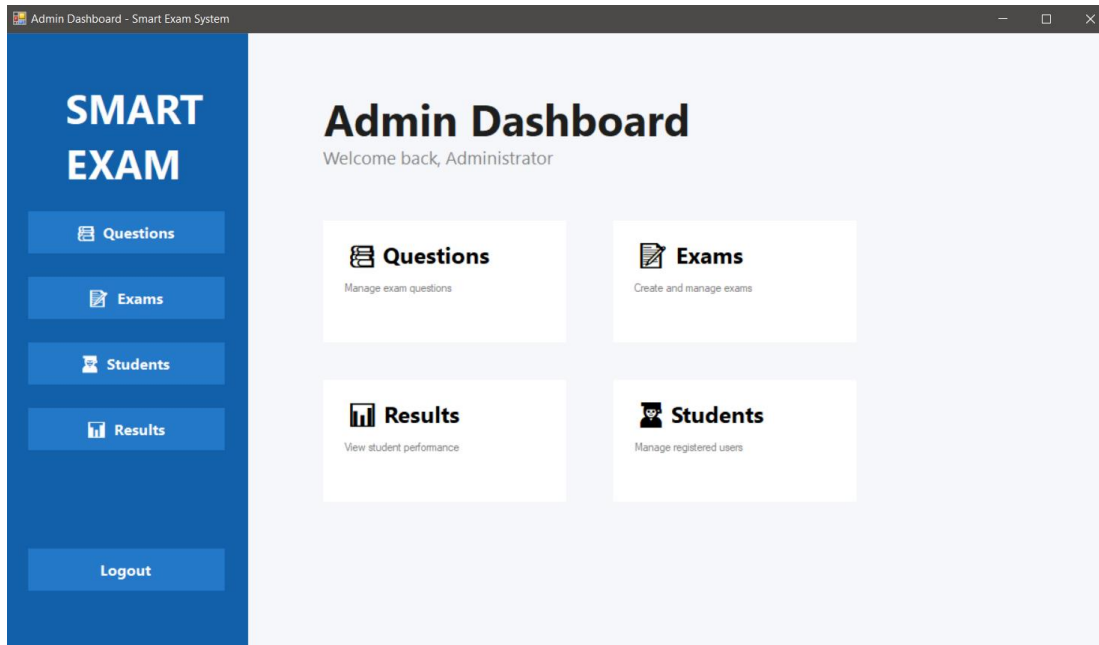
25 1 - 7 of 7

_id	ObjectId	ExamName String	StudentName String	StudentEmail String	StudentRoll String
1	ObjectId('6a33da6500290e4...')	"Entrepreneurship"	"Alsa Javed"	"alsa@gmail.com"	"NIMS-2026-ST"
2	ObjectId('6a33da600290e4...')	"Artificial Intelligence ..."	"Alsa Javed"	"alsa@gmail.com"	"NIMS-2026-ST"
3	ObjectId('6a34b182181835d...')	"Artificial Intelligence ..."	"Alsa Javed"	"NIMS-2026-ST"	"alsa@gmail.com"
4	ObjectId('6a34ef145f3d4f2...')	"Entrepreneurship"	"Jaffar Javed"	"NIMS-2026-ST"	"jaffar@gmail.com"
5	ObjectId('6a34ef685f3d4f2...')	"Artificial Intelligence ..."	"Jaffar Javed"	"NIMS-2026-ST"	"jaffar@gmail.com"
6	ObjectId('6a34f2475f3d4f2...')	"استراتيجيات"	"Jaffar Javed"	"NIMS-2026-ST"	"jaffar@gmail.com"
7	ObjectId('6a362686a06d6d8...')	"C# (CSharp)"	"Alsa Javed"	"NIMS-2026-ST"	"alsa@gmail.com"

f. Exam Module

7.7 Admin Dashboard

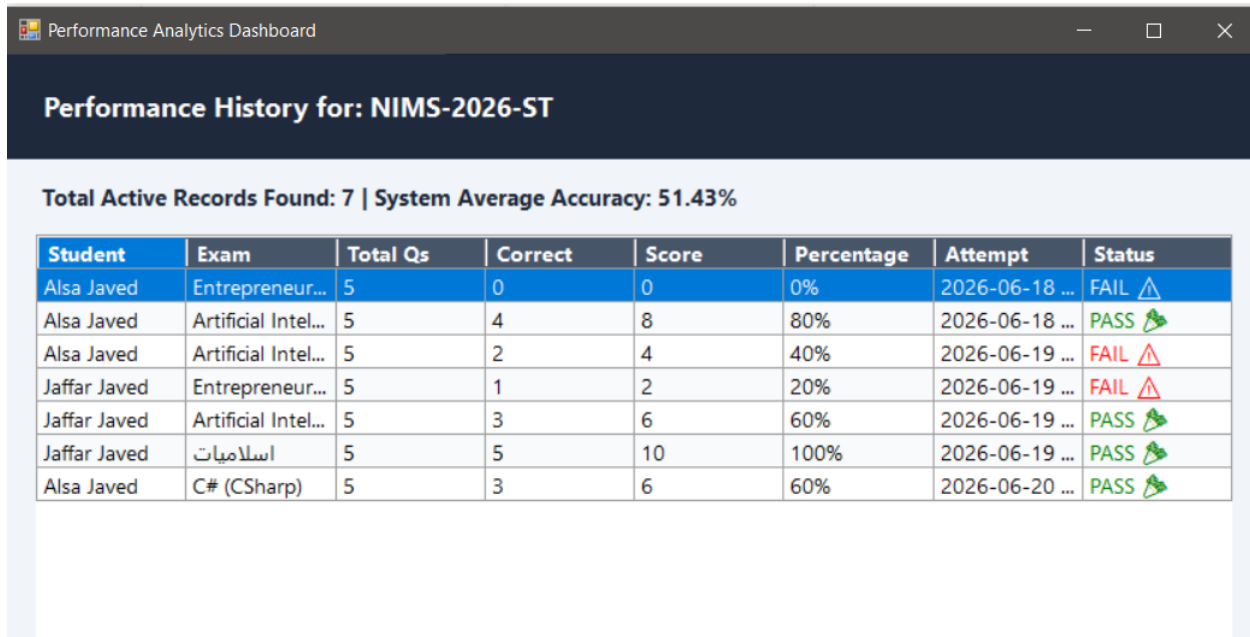
Provides administrators with a centralized view to monitor students, exams, and questions. Acts as the entry point to the Performance Matrix and other management tools.



g. Admin Dashboard

7.8 Admin Performance Matrix Module

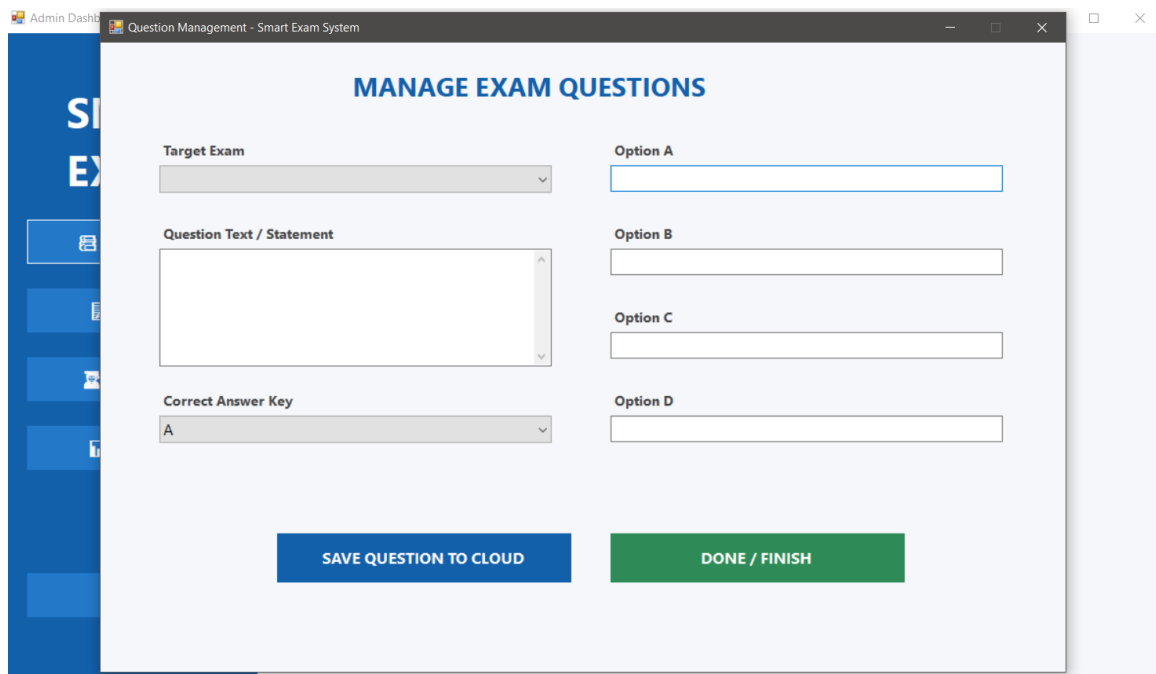
Fetches historical and active exam results from the MongoDB cloud server and displays them in a live grid. Each row is evaluated at runtime and labeled with a bold, color-coded marker — Green for Pass (percentage $\geq 40\%$) and Red for Fail (percentage $< 40\%$).



Performance History for: NIMS-2026-ST

Total Active Records Found: 7 | System Average Accuracy: 51.43%

Student	Exam	Total Qs	Correct	Score	Percentage	Attempt	Status
Alsa Javed	Entrepreneur...	5	0	0	0%	2026-06-18 ...	FAIL
Alsa Javed	Artificial Intel...	5	4	8	80%	2026-06-18 ...	PASS
Alsa Javed	Artificial Intel...	5	2	4	40%	2026-06-19 ...	FAIL
Jaffar Javed	Entrepreneur...	5	1	2	20%	2026-06-19 ...	FAIL
Jaffar Javed	Artificial Intel...	5	3	6	60%	2026-06-19 ...	PASS
Jaffar Javed	اسلاميات	5	5	10	100%	2026-06-19 ...	PASS
Alsa Javed	C# (CSharp)	5	3	6	60%	2026-06-20 ...	PASS



MANAGE EXAM QUESTIONS

Target Exam:

Question Text / Statement:

Correct Answer Key:

Option A:

Option B:

Option C:

Option D:

SAVE QUESTION TO CLOUD **DONE / FINISH**

CREATE MANAGE EXAMS

Exam Name

Duration (Minutes)

Total Questions

CREATE EXAM

Id	ExamName	TimeDuration	TotalQuestions
6a33d31330677...	C# (CSharp)	2	5
6a33d4bf30677e...	Artificial Intelligen...	2	5
6a33d64630677...	Natural Languag...	2	5
6a33d80c30677...	Entrepreneurship	2	5
6a33d89530677...	Data Mining	2	5
6a34f07e5f3d42...	السلاميات	1	5

Admin Dashboard - Smart Exam System

SMART EXAM

Questions

Exams

Students

Results

Logout

Admin Dashboard

Welcome back, Administrator

Roll Number	Full Name	Date of Birth	Email Address	Role
	Alsa Javed		alsa@gmail.com	Student
	Maryam		may@gmail.com	Student
NIMS-2026-ST244	Aleena Jadoon	2021-05-19	aleena@gmail.com	Student
NIMS-2026-ST511	Jaffar Javed	2010-02-13	jaffar@gmail.com	Student

Exam Results Center - Admin Panel

STUDENT EXAM PERFORMANCE MATRIX

ResultID	Roll Number	Student Name	Email Address	Exam Name	Total Qs	Correct	Marks	Percentage	Status	Date
	NIMS-202...	Alsa Javed	alsa@gm...	Entrepren...	5	0	0	0	Fail	6/18/202...
	NIMS-202...	Alsa Javed	alsa@gm...	Artificial In...	5	4	8	80	Pass	6/18/202...
	alsa@gma...	Alsa Javed	NIMS-202...	Artificial In...	5	2	4	40	Pass	6/19/202...
0	jaffar@gm...	Jaffar Javed	NIMS-202...	Entrepren...	5	1	2	20	Fail	6/19/202...
0	jaffar@gm...	Jaffar Javed	NIMS-202...	Artificial In...	5	3	6	60	Pass	6/19/202...
0	jaffar@gm...	Jaffar Javed	NIMS-202...	السلاميات	5	5	10	100	Pass	6/19/202...
0	alsa@gma...	Alsa Javed	NIMS-202...	C# (CSharp)	5	3	6	60	Pass	6/20/202...

h. Admin Performance Matrix

8. Database Design

The database design of the Smart Exam System consists of two main MongoDB collections:

8.1 Users Collection

Stores all user accounts, including students and admins.

Field	Data Type	Description
Name	String	Full name of the user
Email	String	Email address used for login
Password	String	Account password
Role	String	Student, Teacher, or Admin
RollNumber	String	Auto-generated unique roll number
DateOfBirth	DateTime	Used for age-limit validation
Semester	String	Default or current semester

i. Users Collection Schema

8.2 Results Collection

Stores all exam results submitted by students.

Field	Data Type	Description
StudentRoll	String	Roll number of the student
StudentName	String	Name of the student
ExamName	String	Name of the exam attempted
TotalQuestions	Int	Total number of questions
CorrectAnswers	Int	Number of correct answers
Marks	Double	Marks obtained
Percentage	Double	Calculated percentage score
Status	String	Pass or Fail
Date	DateTime	Date and time of the exam

j. Results Collection Schema

9. Implementation Details

The following logic represents the core operations of the Smart Exam System using the MongoDB C# driver.

9.1 Registration Validation

```
if (age < 15)
    MessageBox.Show("You must be at least 15 years old to register.");
else if (!email.Contains("@") || !email.Contains(".com"))
    MessageBox.Show("Please enter a valid email address.");
else if (password != confirmPassword)
    MessageBox.Show("Passwords do not match.");
else
    usersCollection.InsertOne(newUser);
```

9.2 Automatic Roll Number Generation

```
string rollNumber = $"NIMS-{DateTime.Now.Year}-ST{new Random().Next(1000, 9999)}";
newUser.RollNumber = rollNumber;
newUser.Semester = "1st Semester";
```

9.3 Result Calculation

```
double percentage = (correctAnswers / (double)totalQuestions) * 100;
string status = percentage >= 40 ? "Pass" : "Fail";
resultsCollection.InsertOne(new ResultModel {
    StudentRoll = roll, Percentage = percentage, Status = status
});
```

9.4 Fetching the Performance Matrix

```
var results = resultsCollection.Find(_ => true).ToList();
dataGridView.DataSource = results;
```

10. Testing

The following test cases were performed to verify that all system functionalities work correctly and produce the expected results.

Feature	Test Case	Expected Result
Login	Valid credentials	Login successful, redirected by role
Login	Wrong password	Error message displayed
Registration	Age below 15	Age-limit validation error shown
Registration	Invalid email format	Email format error shown
Registration	Passwords do not match	Mismatch alert shown before insertion
Registration	Valid new student	Roll number auto-generated, registered successfully
Exam Submission	Valid attempt	Result saved with correct percentage
Performance Matrix	Percentage $\geq 40\%$	Row marked Pass in green
Performance Matrix	Percentage $< 40\%$	Row marked Fail in red
Password Toggle	Click Show/Hide icon	Password visibility toggles correctly

11. Conclusion

The Smart Exam System successfully delivers a complete examination management solution for students and academic institutions. With role-based access, automated student onboarding, real-time result calculation, and a centralized performance matrix, the system covers the full examination lifecycle efficiently.

Built on C# .NET WinForms and MongoDB Atlas, it demonstrates practical implementation of cloud database connectivity, dynamic UI rendering, and secure multi-role user management. The inclusion of security features such as age-limit validation, email verification, and password masking reflects a strong commitment to data integrity and user account safety.

Overall, this project serves as a valuable and professional solution for automating the examination process in a transparent, scalable, and error-free manner.